

Week 4: Scope locked, Network 1 built

Repo: github.com/ltsabhirajurs/Trust_DTI · Decisions: DECISIONS.md D-011–D-019 · Table: [data/processed/network1_v1.parquet](#)

DONE

Locked the Week 2 scope rules from real data (9 decisions), then built Network 1: the clean, labelled, leakage-checked drug–protein table with four train/test splits.

KEY RESULT

748k active and 1.16M inactive pairs (1.91M total) over 2,312 proteins and 1.01M compounds; all four splits pass the automated leakage check.

NEXT

Stage 3 — embed compounds (ChemBERTa) and proteins (ESM-2) on a Kaggle GPU, cache the vectors, and build classical baselines.

1 Where we are in the plan

Stage 1 (Data) finished in the previous reporting period. This week completed **project Week 2** (lock the dataset rules) and **project Week 3** (build Network 1) — Stages 1 and 2 of the pipeline are now done. The model itself (Stages 3–4) is next and needs a GPU, which is why it is scheduled on Kaggle.

How Network 1 is built: 8.3M messy records — one clean labelled table

Stage 2 of the pipeline. Every step is a committed, re-runnable script in `src/network/`. The rules each step applies were locked in Week 2 from real data (DECISIONS.md D-011 to D-019).

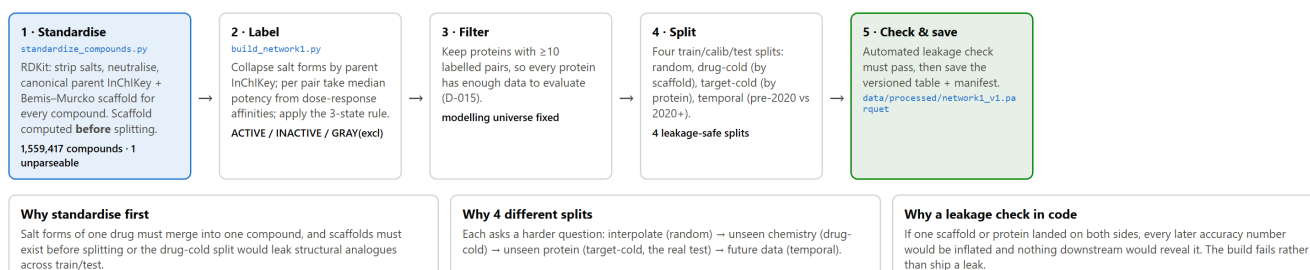


Figure 1. The five steps that turn 8.3M raw measurements into one clean labelled table. Each is a committed script in `src/network/`.

2 Locking the scope from evidence (project Week 2)

Before building anything, we had to decide what the data actually means: which measurements to trust, where the line between "active" and "inactive" sits, and how to split the data fairly. We made each decision by running an analysis on all 8.3M records, then recording the choice and the numbers behind it. The analysis script is `src/ingest/analyze_scope_week2.py`; the decisions are DECISIONS.md D-011 to D-019.

Week 2 scope decisions — locked from real evidence, not guessed

Each was decided by running an analysis on all 8.3M records, then recording the choice, the reason and the numbers in DECISIONS.md. These define exactly what Network 1 contains.

ID	DECISION	EVIDENCE THAT DROVE IT
D-011	Label only from dose-response affinities (IC50/Ki/Kd/EC50); qHTS as inactive evidence only	Dose-response = 2.77M rows; restricting to them removes the PubChem distortion (median pChEMBL 6.85 → 6.91 without PubChem)
D-012	ACTIVE ≥ 6.5, INACTIVE ≤ 5.0, gray band excluded	Gives 754k ACTIVE + 1.16M INACTIVE; the 1.5-log gray buffer keeps labels clean for calibration
D-013	No positive-unlabelled fallback needed	1.16M real measured negatives exist (762k from "not active" comments)
D-014	Compound identity = salt-stripped parent InChIKey; don't merge stereo/isotope variants	Scaffold grouping already stops those leaking; merging would conflate molecules with different activity
D-015	Keep proteins with ≥10 labelled pairs	2,313 proteins (2,008 with both classes), loses only ~2k of 1.91M pairs
D-016	Temporal split at year 2020	861k train / 396k test (31.5% test) among dated pairs
D-017	ChEMBL-only for Network 1 v1	BindingDB mostly duplicates ChEMBL (Week 1 finding)
D-018, D-019	Median aggregation + keep disagreement; four leakage-safe splits, seed 42	23% of repeat-measured pairs disagree by ≥1 log unit, so spread is kept and flagged

Figure 2. The nine scope decisions and the evidence that drove each one.

The single most useful finding: restricting labelling to **dose-response affinity** measurements (IC50/Ki/Kd/EC50) nearly removes the PubChem screening distortion found in Week 1 — the median potency barely moves when PubChem is excluded (6.85 → 6.91), so we did not need complicated per-family corrections. We also confirmed we have abundant real negatives (1.16M measured-inactive pairs), so the positive-unlabelled fallback is not needed.

3 How each pair gets its label

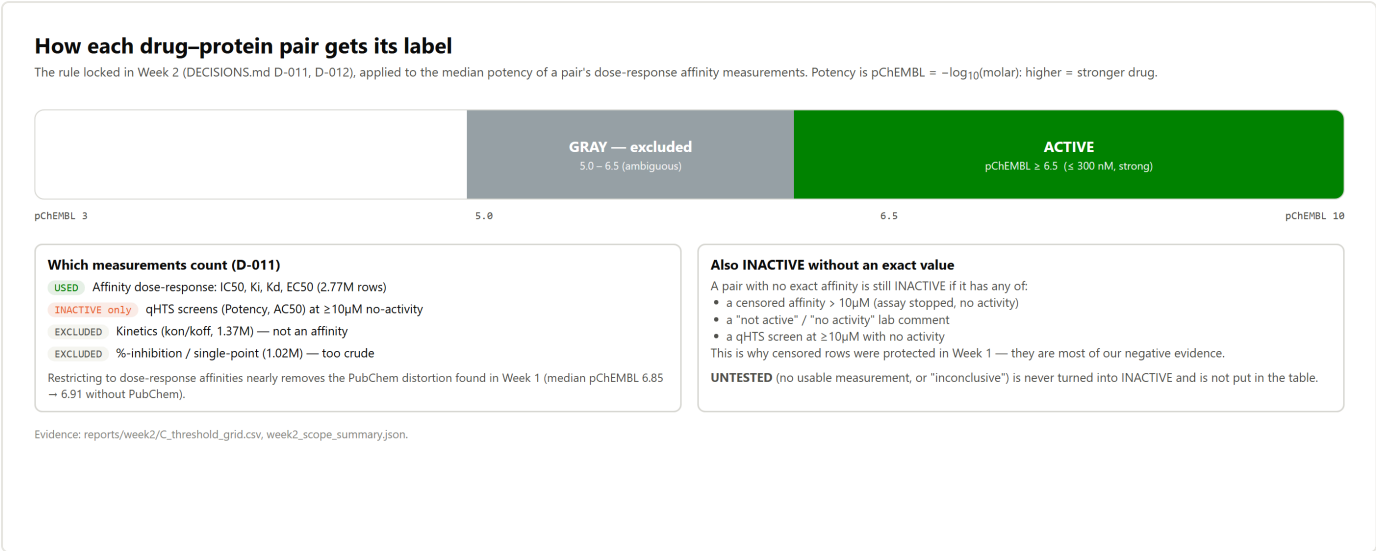


Figure 3. The labelling rule, applied to the median potency of each pair's dose-response measurements. The ambiguous 5.0–6.5 band is excluded to keep labels clean; "untested" is never turned into "inactive".

4 What Network 1 contains

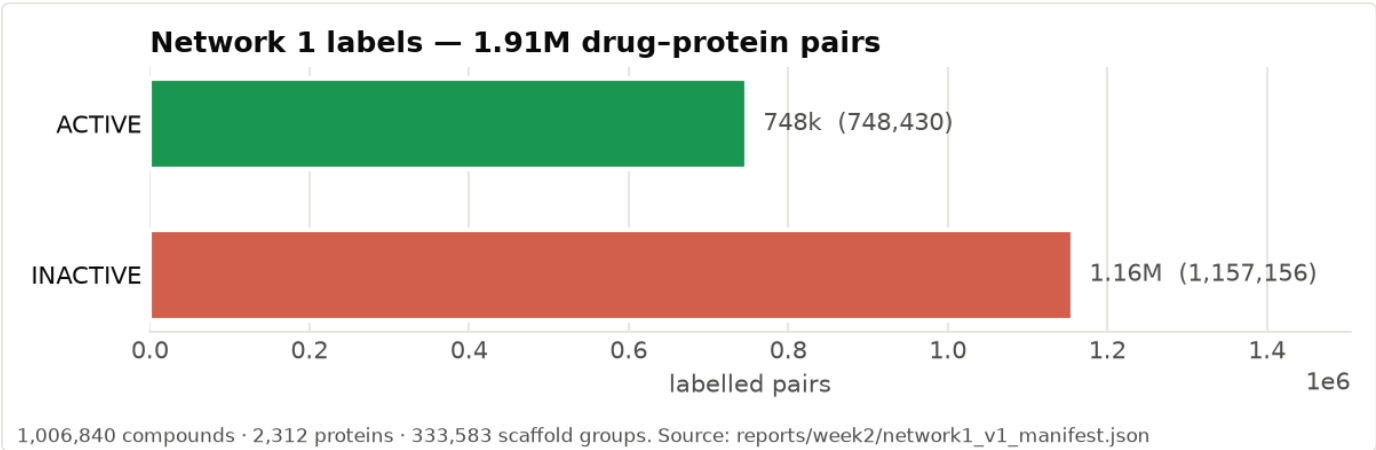


Figure 4. The finished table: labelled drug–protein pairs, with compound, protein and scaffold counts.

COLUMN	WHAT IT HOLDS
pik, accession	The compound (salt-stripped parent InChIKey) and protein (UniProt) — the pair's identity
label	ACTIVE or INACTIVE
median_p, iqr, high_disagreement	Aggregated potency, its spread, and a flag where repeats disagree by ≥ 1 log unit
scaffold, scaffold_group	The compound's core ring system, used for the drug-cold split
year	Earliest measurement year, used for the temporal split
split_random / drugcold / targetcold / temporal	train / calib / test assignment for each of the four splits
provenance	"measured" — so predicted edges (added later) can never be confused with measured ones

5 The four splits and the leakage check

The four train / test splits, and the question each one answers

Every split also carves a small **calibration** slice from its own training data, kept for the Week 7 uncertainty step. A leakage check confirms nothing held out for testing also appears in training.

SPLIT	WHAT IS HELD OUT FOR TESTING	QUESTION IT ANSWERS	WHY IT MATTERS
Random	A random 15% of pairs	Can the model interpolate within chemistry and biology it has already seen?	The easiest test — a sanity floor. Good scores here alone prove little.
Drug-cold	Whole scaffolds (core ring systems)	Does it work on chemical structures it has never seen?	Grouping by scaffold stops near-identical molecules leaking across the split.
Target-cold primary	Whole proteins	Does it work on a protein it has never seen in training?	This is what real drug repurposing actually is — so it is the headline test.
Temporal	Everything measured in 2020 or later	Does it work on data from the future, trained only on the past?	The closest thing to a genuine prospective test. Limited by the 40% of records with no date.

Fixed random seed 42, recorded in the manifest, so the exact same splits regenerate on any machine. Rules: DECISIONS.md D-019.

Figure 5. Each split holds out something different and answers a harder question than the last.

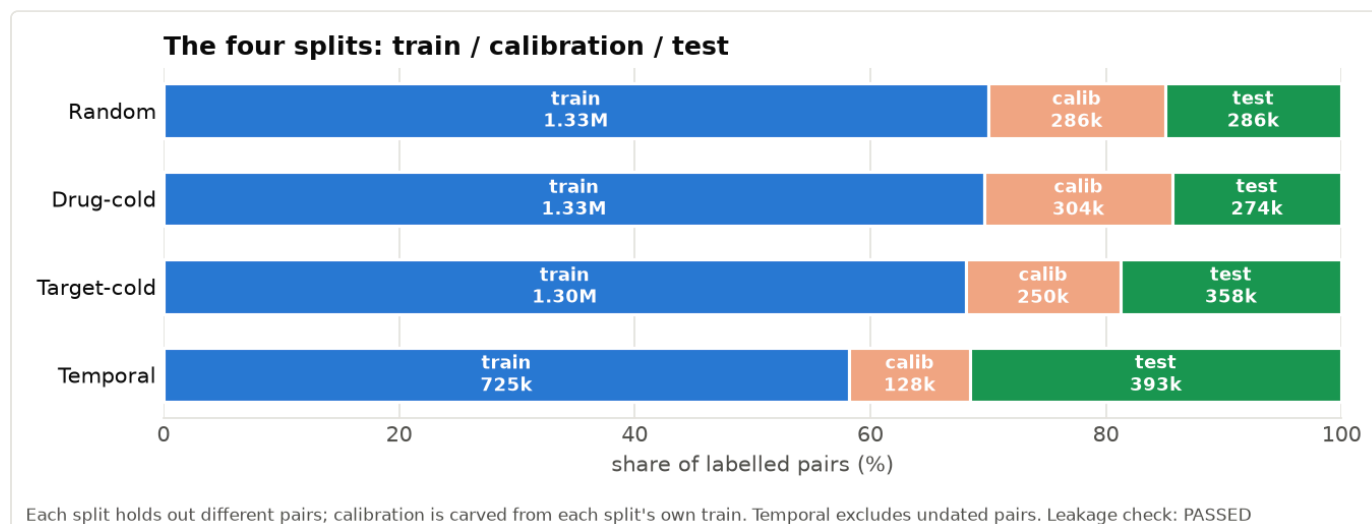


Figure 6. How the labelled pairs divide into train / calibration / test for each split. The build fails unless the leakage check passes.

The leakage check is code that **fails the build** if any scaffold appears in both the drug-cold train and test sets, or any protein in both the target-cold sets, or if the temporal calibration data leaks past the 2020 cutoff. For this build it **passed**.

6 Roadblocks this week

- **No local GPU** (confirmed): Stage 3 embeddings (1.56M compounds, ~5,700 proteins) can't run on the laptop, so they are scheduled on a Kaggle GPU. Not a blocker for Network 1.
- **Memory limits**: the first Network 1 build was silently killed by the OS — holding all 8.3M rows with their free-text comment column plus Python set operations exceeded 16 GB RAM. Fixed by dropping the text column early, using category types, and replacing Python sets with vectorised pandas joins. Build memory dropped from 2.3 GB (killed) to 1.6 GB.
- Two further build bugs caught and fixed: a read-only-array shuffle, and a non-serialisable missing-value key in the manifest.
- **Measurement disagreement**: 23% of repeat-measured pairs disagree by ≥ 1 log unit, so we aggregate by median and flag high-disagreement pairs rather than trusting any single value.

7 Next week (Stage 3)

- Embed every Network 1 compound with ChemBERTa and every protein with ESM-2 (frozen, on Kaggle GPU); cache the vectors keyed by identifier.
- Handle the giant proteins (longest is 34,350 residues vs ESM-2's ~1,024 limit) by truncation.
- Build classical fingerprint baselines (logistic regression, XGBoost) — the floor the Stage 4 model must beat.
- Sanity-check that similar compounds and same-family proteins cluster in the vector space.

Supporting files: W4_01–W4_06 (figures), references.md, WEEK04_FORM_ENTRIES.md. All figures regenerate from committed data.
Reproduce the table: `python -m src.network.standardize_compounds` then `python -m src.network.build_network1`.